



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Trimester: Spring 2026 Marks: 40 Time: 2 Hours

Code: CSE 3411 Course Title: System Analysis & Design

“Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.”

Answer all the following questions:

Question 1:

[CO3]

[18]

Consider the following scenario:

FIFA is deploying a centralized system called "**MatchOps26**" to manage stadium operations during the FIFA World Cup 2026 across multiple venues in the USA, Canada, and Mexico. The system connects the FIFA app, ticketing kiosks, gate scanners, and stadium display screens at each venue to a local server, which synchronizes with FIFA's central cloud server for global match data and fan records.

The system supports three types of users:

Fans use the FIFA app to view ticket details and receive entry notifications. To buy a ticket, a fan selects a match and seat through the app or a ticketing kiosk, and the system redirects the request to a third-party Payment Gateway. At the stadium, fans scan their digital tickets at gate scanners connected to the venue's local server. **Stadium Staff** monitor gate scanner status and manually verify flagged tickets at the help desk. They use tablets connected to the venue's local Wi-Fi network and local server. **FIFA Administrators** manage match schedules and the ticket database from FIFA's central operations center. They can also remotely lock down gate scanners at any venue during emergencies.

Each venue's local server validates ticket entries and periodically syncs logs with the central cloud server over a secure internet connection. The central cloud server hosts the main ticket and fan profile databases and communicates with the external Payment Gateway server to process ticket purchases.

A digital ticket is created in the **PENDING_PAYMENT** state when a fan selects a seat and proceeds to checkout. The central server sends the payment request to the Payment Gateway. If payment fails or is declined, the ticket moves to **PAYMENT_FAILED** and the seat is released for sale. If payment succeeds, the ticket transitions to **ISSUED** and is sent to the fan's app. When the fan scans the ticket at a gate scanner, the ticket enters the **VALIDATING** state while the system checks it against the central database. If the ticket is duplicate, tampered with, or reported stolen, it moves to **FLAGGED**, and the fan is directed to the help desk for manual verification. If validation succeeds, the ticket moves to **VALIDATED** and the turnstile opens. After the fan passes through, the ticket enters the **USED** state and cannot be scanned again for that match. If a ticket remains unscanned until a specified time after the match begins, it automatically transitions to **EXPIRED**.

- A) Draw the Use Case Diagram of the MatchOps26 system.** [4]
- B) Draw the Class Diagram and 2 major class CRC card of the MatchOps26 system.** [4 + 2]
- C) Draw the Sequence Diagram for the process of purchasing a ticket and gaining entry to a stadium using the "MatchOps26" system.** [4]
- D) Create a State Diagram that depicts the life-cycle of a digital ticket in the "MatchOps26" system.** [4]

Question 2: [CO4] [5]

A) A software company estimates that a project will require 8 months to complete. However, the client demands delivery within 4 months without reducing features or increasing resources. Is the project schedule feasible? Justify your answer. [2]

B) A software company is considering developing an **Online Course Management System** for a university. The project requires an initial investment of **BDT 15 million**. The university estimates that the system will save approximately **BDT 2 million per year** in administrative costs and generate an additional **BDT 1 million per year** through online certification services. The system is technically feasible, can be completed within the required timeline, and users are willing to adopt it. Based on the feasibility information provided, would you recommend proceeding with the project? Identify the most critical feasibility factor and justify your answer. [3]

Question 3: [CO3] [12]

A) A software company is developing a **Hospital Management System**. During an SRS review meeting, the project manager notices that one requirement states that *patients can book appointments online*, but there is no mention of how appointment cancellations or rescheduling should be handled. The developers also find that the SRS document is difficult to update because the same requirement appears in several different sections, meaning a change in one place requires manual changes throughout the document. Furthermore, when asked where the appointment-booking requirement originated, the team cannot determine whether it came from hospital management, doctors, or patients because no source references have been documented. Identify any **two missing characteristics of a good SRS** in the above scenario and explain why they are missing. [2+2]

B) Consider the same **Hospital Management System** described in Question(A). Classify each requirement as either a **Functional Requirement (FR)** or a **Non-Functional Requirement (NFR)**. [3]

- Patients shall be able to book appointments online.
- Doctors shall be able to view patient medical records.
- The system shall respond to appointment booking requests within 3 seconds.
- The system should be available 24 hours a day with 99.9% uptime.
- Receptionists shall be able to generate daily appointment reports.
- All patient data shall be encrypted during storage and transmission.

C) Design the landing page of *ShareAid*, a platform for donating unused household items to people in need. Include key features for donors, recipients, and community impact [3]

d) Analyze your proposed design from Question (c). Explain exactly how you applied at least three of the following UI design concepts to ensure the application is intuitive, efficient, and error-free: Affordance & Signifiers, Feedback, Constraints, Consistency. [2]

Question 4: [CO3] [5]

Write down an SRS document and present it as a group having the following parts on your software project work covering: Introduction, System Study, System Analysis, Feature list fixation with Functional and Non-functional requirements, Different Feasibility Analysis, System design (UML diagrams), UI design through Figma software, implementation plan etc. (**no need to answer this question in the examination**, it has already been evaluated in classroom activities). [5]